12 **(a)** Show that $\cos x = \sin\left(x + \frac{\pi}{2}\right)$. **[2]**

(b) Hence show that $\sin x \approx \frac{16x(\pi - x)}{5\pi^2 - 4x(\pi - x)}$ gives the approximation $\cos x \approx \frac{\pi^2 - 4x^2}{\pi^2 + x^2}$, as stated in line 31. **[3]**

12(a)

12(b)

(answer space continued on next page)

12(b) **(continued)**